SI: Sulfur_X: A model of sulfur degassing during magma ascent

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Introduction

This document includes step-by-step instructions for using Sulfur_X (text S1), with a table of example inputs to reproduce the model results in the main text (Table S1) and a table of experiments used for model calibration (Table S2); a brief introduction of the Inverse Iterative Least Squares Method used for regression (Text S2), and eight supplementary figures.

S1. Instructions for using Sulfur_X

Sulfur_X is written in Python and can be run in python 3.9 or any version higher. It requires packages of Pandas, Numpy and Scipy. The package includes one main function script, several other scripts of different classes called by the main function, and two example csv files for inputting melt inclusion data for comparison. All of these files should be kept in the same folder, and none of the names should be changed in any python script file.

To run the program, **open the main function script named "main_Fuego.py", change the basic inputs and the advanced parameters accordingly and run.** Melt composition can be modified in the python script named "**melt_composition.py**". Parameters for low pressure (<20 MPa) degassing (INT and BAR) can be changed in the python script named "**degassingrun.py**". The example input parameters that will reproduce the results for Fuego and Mauna Kea in the main text of this paper are listed in Table 1.

1.1.Basic input

temperature: Magma temperature in °*C.* Degassing is modeled isothermally.

delta_FMQ: fO_2 relative to FMQ buffer (Frost, 1991). If the redox evolution is enabled, this is the initial fO_2 at the initial pressure. If the redox evolution is disabled, the

degassing run will be run along this buffer. The initial Fe^{3+}/Fe_T in the melt is calculated based on this input (Kress and Carmichael, 1991)

H2O_initial: Initial water concentration in the melt in wt.%.

CO2_initial: Initial carbon dioxide concentration in the melt in ppm.

S_initial: Initial total sulfur concentration in the melt in ppm.

choice: choice of crystallization or not. If equals 1, crystallization is enabled; if equals 0, crystallization is disabled.

COH_model: choice of COH degassing model. If equals 1, VolatileCalc (Newman and Lowenstern, 2002); if equals 0, Iacono-Marziano model (Iacono-Marziano et al., 2012). We recommend using VolatileCalc for low water (<1wt.%) system.

fO_2_tracker: choice of redox change along S degassing or not. If equals 1, fO_2 changes by electron exchange between S and Fe in the melt and the co-existing vapor; if equals 0, fO_2 is always buffered at the initial delta FMQ.

monte_carlo: Optional mote-carlo simulation for S degassing error estimate. If equals 1, monte carlo simulation will be performed after the first degassing run, and input the number of simulations as "*m_run*"; if equals 0, the program finishes after one degassing run. *monte_name*: Name (input as str and ends in .csv) of the output csv file of S concentration in the melt (in ppm) along decompression of each simulation, the mean, standard deviation and variance of S (in ppm) at each pressure from multiple simulations. This file will be rewritten each time the code is run (even if the monte-carlo simulation is turned off). In order to save the results, either change the name or save the output file under a different name before the next run.
l: The total number of pressure steps between the initial pressure and 1bar. We recommend this number to be between 300-500. Too large or too small pressure steps can both cause unexpected excursions in the model results.
m: The total number of the degassing runs during decompression. $0 \leq m \leq l$.

sulfide: Sulfide composition, S, Fe, Ni, Cu and O, in wt.%. This is used to calculate the sulfur contents at sulfide saturation (SCSS) using Smythe et al. (2017) model. The SCSS calculated here is only for the reference. Sulfur_X currently does not consider sulfide saturation along degassing.

1.2. Advanced parameters

Changes in any of the following parameters could have significant impacts on the model results. Please change the following parameters with caution and note the changes if the results are used.

1.2.1 key parameters for redox and crystallization calculation in ***main_Fuego.py***

S_Fe_choice: choice of S speciation model. If equals 0, Nash model is employed (Nash et al., 2019); if equals 1, O'Neill and Mavrogenes (2022) model is employed; if equals 100, Muth model is employed (Muth and Wallace, 2021); if equals any other float number, revised Muth model is employed and the input float number is the last constant in the Muth model. The choice of this model is the key for modeling sulfur degassing and fO_2 evolution. We recommend users choose the model that fits the *S-Fe* speciation measurements in the melt for the volcanic system of interest.

Sigma: The tolerance of the calculated $\log_{10}fO_2$. When change in $\log_{10}fO_2$ is smaller than sigma, the iterations searching for fO_2 satisfying all the conditions stops (more details in the text). For the example runs, sigma equals to 0.05. If sigma is too small or too large, it may cause large excursions or even false fO_2 calculations in the run.

If the crystallization is enabled, H_2O (in wt.%)-melt fraction relation is specified using H_2O (wt.%)-K_2O (wt.%) correlation ($K_2O = a*H_2O + b$), assuming K_2O is perfectly incompatible. The following two parameters, *slop_h2o* and *constant_h2o* represent a and b, respectively.
slop_h2o: slop of H_2O (wt.%) in K_2O (wt.%) as a function of H_2O.
constant_h2o: constant in K_2O (wt.%) as a function of H_2O.
Change the slope and constant based on the H2O-K2O relation of the volcanic system of interest. If the system is assumed to be similar to Fuego, or crystallization is disabled, leave them unchanged.

1.2.2. Melt composition in ***melt_composition.py***
Melt composition is used to calculate the sulfur partition coefficients and COH degassing if the Iacono-Marziano model is used. **Initial melt composition and the change as a function of melt fraction can be changed in the python script named "melt_composition.py".** If crystallization is enabled, the default input of the melt composition change as a function of melt fraction is based on Fuego magma (Lloyd et al., 2013; Rasmussen et al., 2020). If crystallization is disabled, the default input of the melt composition is an averaged composition for Hawaiian magma (Moussallam et al., 2016; Brounce et al., 2017).
*1.2.3. Low pressure degassing (<25 MPa) in "****degassingrun.py****"*
Since there is very little experimental information at pressures lower than 25 MPa except for the H-free experiments from Nash et al. (2019) and O'Neill (2002), the predicted sulfur partition coefficients increase exponentially at this very low-pressure range and large uncertainties are involved. This might cause large excursions of the model results at low pressure, especially in the fO_2 calculation. Therefore, Sulfur-X has the option to impose an arbitrary incremental increase in the combined molar partition coefficients of sulfur (*INT*) at each pressure step at low pressure (*BAR in MPa*, should be smaller than 20 MPa). *INT* and *BAR* can be changed in the python file named "**degassingrun.py**". If there are large excursions in the fO_2 calculation in the last a few pressure steps, we recommend the users to play with these two parameters to get the best fO_2 results.
1.3 Input of melt inclusion data for comparison in ***main_Fuego.py***
If you'd like to compare your model results to H_2O-CO_2-S-$Fe^{3+/}Fe_T$-$S^{6+/}S_T$ data measured in the melt inclusions, you can use either of provided csv files (named "Fuego.csv" and

"Hawaii.csv") as a templet and change the read-in csv name in **main_Fuego.py**. Please do not change any of the column names in the csv templet.
mi_name: name (input as str) of the csv file containing melt inclusion data.
1.4 *Output file and plots in* ***main_Fuego.py***
All the results in the dataframe "df_results" are output in a csv file. The output file name should be changed for each run. Otherwise, the result file will be rewritten.
output_name: name (input as str) of the csv file for the model results.
Result figures similar to Figure 6, 7, 8 and 9 in the main text will be plotted at the end of the run (results of monte-carlo simulations are not included). This part can be modified as needed.
In addition to the key outputs discussed in the main text, the output file also includes calculated sulfur contents at sulfide saturation (SCSS, Smythe et al., 2017), sulfate contents at anhydrite saturation (SCAS, Zajacz and Tsay, 2019) and hydrogen fugacity (fH_2, Ohmoto and Kerrick, 1977) along degassing. These values only serve as references if the users are interested, and do not affect the actual degassing calculations. If these results are used, please make sure cite the references listed above accordingly.

Table S.1 Inputs for Fuego and Mauna Kea examples

	Fuego	Mauna Kea
temperature (°C)	1030	1150
delta_FMQ	1.2	0.5
H2O_initial (wt.%)	4.5	0.6
CO2_initial (ppm)	3300	4000
S_initial (ppm)	2650	1500
COH_model	1	0
fO_2_tracker	1	1
l	300	600
m	300	600
S_Fe_choice	1	6.3
sigma	0.01	0.01
slop_h2o	-0.713	-
constant_h2o	3.689	-
INT	-	50
BAR (MPa)	0	5

1.5 Troubleshooting guides
1.5.1 When testing a new dataset, we suggest the user turn off the fO_2 tracker first. This initial run will give a baseline S degassing trend without the impact of evolving fO_2 on partition coefficients.
1.5.2 It is important to check that this initial run has a reasonable $S^{6+}/\Sigma S$ in the melt for the dataset. If the calculated $S^{6+}/\Sigma S$ in the melt appears too high or too low, the user should adjust the *S-Fe* (by changing the parameter *S_Fe_choice*) relation in the inputs until the model runs with a reasonable initial $S^{6+}/\Sigma S$ for the given dataset. Then, the user can turn the fO_2 tracker on to track the fO_2 evolution during degassing. Another way to coarsely decide if the calculated $S^{6+}/\Sigma S$ is reasonable for the dataset is to compare the

measured S concentration to the SCSS and SCAS values. $S^{6+}/\Sigma S$ could be estimated using $SCSS/S_{measured}$. As we showed in our case studies, if the melt is mostly dominated by S^{6+}, S degassing paths are similar among different S-Fe models while the predicted fO_2 varies (e.g. Fuego). If the melt is mostly dominated by S^{2-}, different $S^{6+}/\Sigma S$ in the melt could result in larger differences in S degassing (e.g. Mauna Kea) and predicted fO_2 evolution.

1.5.3 Large excursions (particularly if the fO_2 tracker is enabled) could happen at low pressure degassing due to large partition coefficient and low S in the melt. If *Sigma* , the tolerance of the calculated $\log_{10} fO_2$ (section 1.2.2), is too small or too large, or the pressure step (controlled by parameter *l*) is too large, it may cause large excursions of fO_2 calculations in the run. Therefore, adjusting *Sigma* and *l* can help to stabilize the low pressure run. Alternatively, the user can use the *mont*e function to run multiple times and obtain an average results at low pressure degassing. Lastly, if the low-pressure degassing is not of interest (e.g. in studies of comparing to melt inclusion data), the user can simply terminate the degassing run at higher pressure (by change parameter *m*).

S2. Inverse Iterative Least Squares Method

2.1 Define the inverse problem

This method is explained in detail in Chapter 5 (Menke, 2012b) and Chapter 9 (Menke, 2012a) of the book "Geophysical Data Analysis: Discrete Inverse Theory" by William Menke. Here, we only briefly describe the regression procedure using $kd_{RxnI}^{v/m}$ as an example.

As shown in equation (11) in the main text, $kd_{RxnI}^{v/m}$ can be expressed as

$$ln\left(kdS_{RxnI}^{\frac{vapor}{melt}}\right) = \ln K_{RxnI}^{0} + \frac{-\Delta V\ (P-\ P_0)}{RT} + \frac{-\Delta H_{T_0}}{R}\left(\frac{1}{T}-\frac{1}{T_0}\right) + lnf_{H_2O} - \ln X_{\mathrm{FeO}} - \ln\gamma FeO - \ln\mathrm{P} - \ln\varphi H2S + ln\gamma_{FeS}$$

As discussed in the main text, the unconstrained terms on the right side of the equation (11) are $\ln K_{RxnI}^{0}$ and $ln\gamma_{FeS}$. The former should be a constant, and the latter is related to the mole fraction of FeO and the square of the mole fraction of H_2O in the melt.

Therefore, we can rewrite the $kdS_{RxnI}^{\frac{vapor}{melt}}$ to be:

$$kd_{RxnI}^{v/m\ *} = a + bXFeO + c(XH_2O)^2$$

Where $kd_{RxnI}^{v/m\ *} = ln\left(kdS_{RxnI}^{\frac{vapor}{melt}}\right) - (\frac{-\Delta V\ (P-P_0)}{RT} + \frac{-\Delta H_{T_0}}{R}\left(\frac{1}{T}-\frac{1}{T_0}\right) + lnf_{H_2O} - \ln X_{\mathrm{FeO}} - \ln\gamma FeO - \ln\mathrm{P} - \ln\varphi H2S)$, and a, b and c can be solved using the experiments categorized in reaction I. If there are N experiments for reaction I, we can write N equations as:

$$(kd_{RxnI}^{v/m})_1^* = a + bXFeO_1 + c(XH_2O)_1^2$$
$$(kd_{RxnI}^{v/m})_2^* = a + bXFeO_2 + c(XH_2O)_2^2$$
$$\ldots\ldots$$
$$(kd_{RxnI}^{v/m})_N^* = a + bXFeO_N + c(XH_2O)_N^2$$

These equations can then be arranged as the matrix equation **Gm**= **d**, where:

$$\mathbf{G} = \begin{bmatrix} 1 & XFeO_1 & (XH_2O)_1^2 \\ \cdots & \cdots & \cdots \\ 1 & XFeO_N & (XH_2O)_N^2 \end{bmatrix}, \boldsymbol{m} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \boldsymbol{d} = \begin{bmatrix} (kd_{RxnI}^{v/m})_1^* \\ \cdots \\ (kd_{RxnI}^{v/m})_N^* \end{bmatrix}$$

2.2 Solution of the linear inverse problem by maximum likelihood method

Once the linear inverse problem **Gm** = **d** is defined, the most common way of solving it is the method of least squares. This method seeks model coefficients (**m**) such that the predicted data is as close as possible to the observed data. The solution can be written as follow:

$$m_{leastsquares} = (\boldsymbol{G}^T\boldsymbol{G})^{-1}\boldsymbol{G}^T\boldsymbol{d}$$

This method assumes that all data are uncorrelated with an equal variance, σ_d^2. The covariance of the least squares solution is therefore:

$$[cov\boldsymbol{m}] = \sigma_d^2[\boldsymbol{G}^T\boldsymbol{G}]^{-1}$$

However, sometimes errors are not only introduced by the measurements ($(kd_{RxnI}^{v/m})_i^*$ in our case), but also different variables (XFeO, XH_2O in our case). They are uncorrelated, but different, particularly in the case of partition coefficients measured by different experimental studies. This iterative method solves the linear inverse problem (**Gm** = **d**) with a maximum likelihood approach, which considers the different errors involved in both the measurements and the variables. This procedure seeks for the optimum solutions (**m**) that maximize the probability that the predicted value is actually the observed data ($\mathbf{d^{obs}}$). The distribution of the data can be approximated by a Gaussian distribution and a priori probability density function $p_A(\mathbf{d})$ can be approximated as:

$$p_A(\boldsymbol{d}) \propto \exp\left[-\frac{1}{2}(\boldsymbol{d}-\boldsymbol{d}^{obs})^T[cov\boldsymbol{d}]^{-1}(\boldsymbol{d}-\boldsymbol{d}^{obs})\right]$$

The maximum likelihood point of **d** can be calculated by minimizing the argument of the exponential in $p_A(\boldsymbol{d})$ with the constraint that **Gm - d = 0.**

For the case of $kdS_{RxnI}^{\frac{vapor}{melt}}$, we define a new matrix **x:**

$$\boldsymbol{x} = \begin{bmatrix} [(kd_{RxnI}^{v/m})_i^*]_N \\ [XFeO_i]_N \\ [X^2H_2O_i]_N \\ \begin{bmatrix} a \\ b \\ c \end{bmatrix}_N \end{bmatrix}_{4N+3}$$

Accounting for the uncertainties in the measured $(kd_{RxnI}^{v/m})_i^*$, XFeO and XH_2O, one can write the covariance matrix for **x:**

$$cov\boldsymbol{x} = \begin{bmatrix} \left[\boldsymbol{cov}(kd_{RxnI}^{\frac{v}{m}})^*\right]_{N\times N} & & & \\ & [\boldsymbol{covXFeO}]_{N\times N} & & \\ & \cdots & [\boldsymbol{cov}X^2H_2O]_{N\times N} & \\ & & & \begin{bmatrix} \boldsymbol{a} & & \\ & \boldsymbol{b} & \\ & & \boldsymbol{c} \end{bmatrix}_{3\times 3} \end{bmatrix}$$

where $cov(kd_{RxnI}^{\frac{v}{m}})^*$ = diag(($\sigma^2(kd_{RxnI}^{\frac{v}{m}})^*$)), $cov(XFeO)$ = diag(($\sigma^2 XFeO$)), $cov(X^2H_2O)$ = diag(($\sigma^2 X^2H_2O$)), and σ is the standard deviation of each variable, which are extrapolated from the uncertainties related to the experimental data. The uncertainties related to the experiments include the measured $kdS_{RxnI}^{fl/m}$ and $kdS_{RxnII}^{fl/m}$, experimental fO_2 conditions, and the uncertainties in the microprobe analyses. Reported uncertainties on $kdS_{RxnI}^{v/m}$ and $kdS_{RxnII}^{v/m}$ vary between 5-40% (Zajacz et al., 2012; Zajacz et al., 2013; Zajacz, 2014; Fiege et al., 2015), except that no uncertainties are reported in Moune et al. (2008) and Nash et al. (2019). Therefore, an intermediate uncertainty, 20%, is assigned to $kdS_{RxnI}^{v/m}$ and $kdS_{RxnII}^{v/m}$ in the regressions. The uncertainty on the reported fO_2 conditions is taken as 0.25 $\log_{10} fO_2$ unit. Errors related to the compositional terms (e.g., NBO/T, XFeO, 2Ca-Al, etc.) are propagated from the uncertainties in the electron microprobe analyses reported by each study. If the uncertainties are not reported (Moune et al., 2008), 1% uncertainty is assigned to abundant oxides (SiO_2, Al_2O_3, CaO, MgO, FeO) and 5% to less abundant oxides (TiO_2, Na_2O, K_2O, and P_2O_5).

The prior distribution for **x** is therefore:

$$p_A(\boldsymbol{x}) \propto \exp\left[-\frac{1}{2}(\boldsymbol{x} - \langle \boldsymbol{x} \rangle)^T [cov\boldsymbol{x}]^{-1} (\boldsymbol{x} - \langle \boldsymbol{x} \rangle)\right]$$

Where $\langle \boldsymbol{x} \rangle = \begin{bmatrix} [(kd_{RxnI}^{v/m})_{i,obs}^*]_N \\ [XFeO_{i,obs}]_N \\ [X^2H_2O_{i,obs}]_N \\ \begin{bmatrix} a \\ b \\ c \end{bmatrix}_N \end{bmatrix}_{4N+3}$, containing the observed $(kd_{RxnI}^{v/m})_i^*$, XFeO, X^2H_2O, and the coefficients a, b, and c. To maximize the $p_A(\boldsymbol{x})$, we need to minimize the argument of the exponential in $p_A(\boldsymbol{x})$, $\emptyset = (\boldsymbol{x} - \langle \boldsymbol{x} \rangle)^T [cov\boldsymbol{x}]^{-1} (\boldsymbol{x} - \langle \boldsymbol{x} \rangle)$, with the constraint that

$$f(\boldsymbol{x_i}) = (a + bXFeO_i + c(XH_2O)_i^2) - (kd_{RxnI}^{v/m})_i^* = 0$$

We took the derivative of f(**x**) with respect to each part of **x** to get a gradient matrix $\boldsymbol{F}_{i,j} = \frac{\partial f_i}{\partial x_j}$.

Therefore, a single equation can be derived for **x:**

$$\boldsymbol{x} - \langle \boldsymbol{x} \rangle = [cov\boldsymbol{x}]\boldsymbol{F}^T\{\boldsymbol{F}[cov\boldsymbol{x}]\boldsymbol{F}^T\}^{-1}\{\boldsymbol{F}[\boldsymbol{x} - \langle \boldsymbol{x} \rangle] - f(\boldsymbol{x})\}$$

Using the solution from the linear least squares method as an initial trial solution ($\mathbf{x}_k$, k=0), **x** can be solved iteratively by generating the successive approximations until the solutions converge.

$$\boldsymbol{x}_{k+1} = \langle x \rangle + [cov\boldsymbol{x}]\boldsymbol{F}_k^T\{F_k[cov\boldsymbol{x}]\boldsymbol{F}_k^T\}^{-1}\{\boldsymbol{F}_k[\boldsymbol{x_k} - \langle \boldsymbol{x} \rangle] - f(\boldsymbol{x_k})\}$$

The covariance of the estimated **x** can be written as:

$$cov\boldsymbol{x}_{est} = [cov\boldsymbol{x}]\boldsymbol{F}^T\{\boldsymbol{F}[cov\boldsymbol{x}]\boldsymbol{F}^T\}^{-1}\boldsymbol{F}[cov\boldsymbol{x}]\{[cov\boldsymbol{x}]\boldsymbol{F}^T\{\boldsymbol{F}[cov\boldsymbol{x}]\boldsymbol{F}^T\}^{-1}\boldsymbol{F}\}^T$$

$$cov(\boldsymbol{kd}_{\boldsymbol{RxnI}}^{\boldsymbol{v/m}})_{\boldsymbol{est}}^{*} = covx_{est}(1{:}N, 1{:}N)$$

$$variance(\boldsymbol{kd}_{\boldsymbol{RxnI}}^{\frac{\boldsymbol{v}}{\boldsymbol{m}}})^{*} = diag(cov(\boldsymbol{kd}_{\boldsymbol{RxnI}}^{\boldsymbol{v/m}})_{\boldsymbol{est}}^{*})$$

$$\sigma(95\%) = 2\sqrt[2]{mean(variance((\boldsymbol{kd}_{\boldsymbol{RxnI}}^{\frac{\boldsymbol{v}}{\boldsymbol{m}}})^{*})}$$

Supplementary Figures

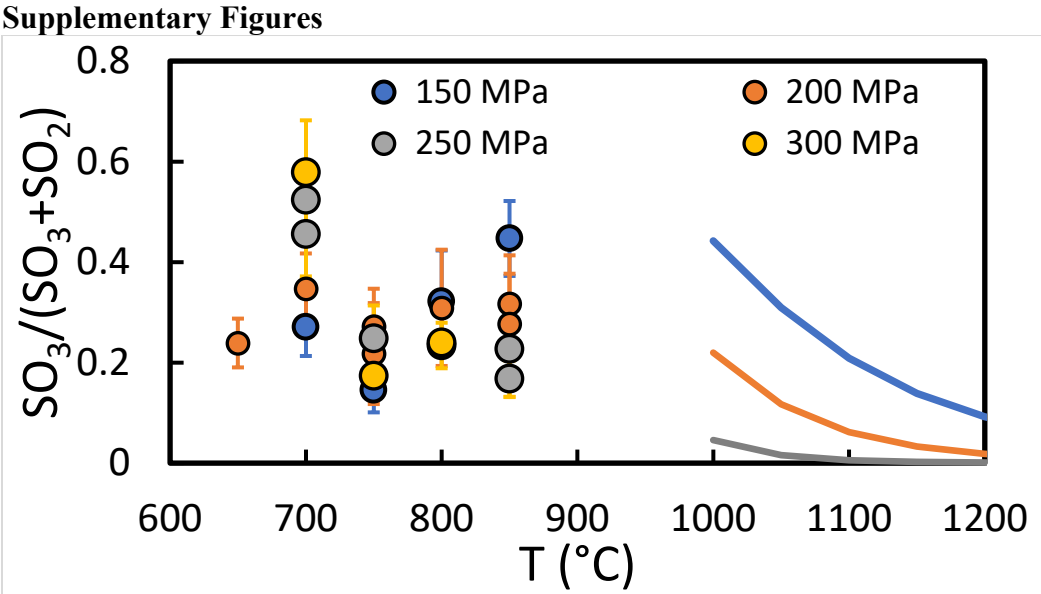

Figure S1. Measured $SO_3/(SO_3+SO_2)$ in fluid vs. temperature constrained by experiments at 150-300 MPa from Binder and Keppler (2011) (filled circles). Curves are extrapolated $SO_3/(SO_3+SO_2)$ in the fluid to a temperature range relevant to basalts and andesites using

the parameters constrained by Binder and Keppler (2011) at 150 MPa (blue), 200 MPa (orange), and 250 MPa (grey).

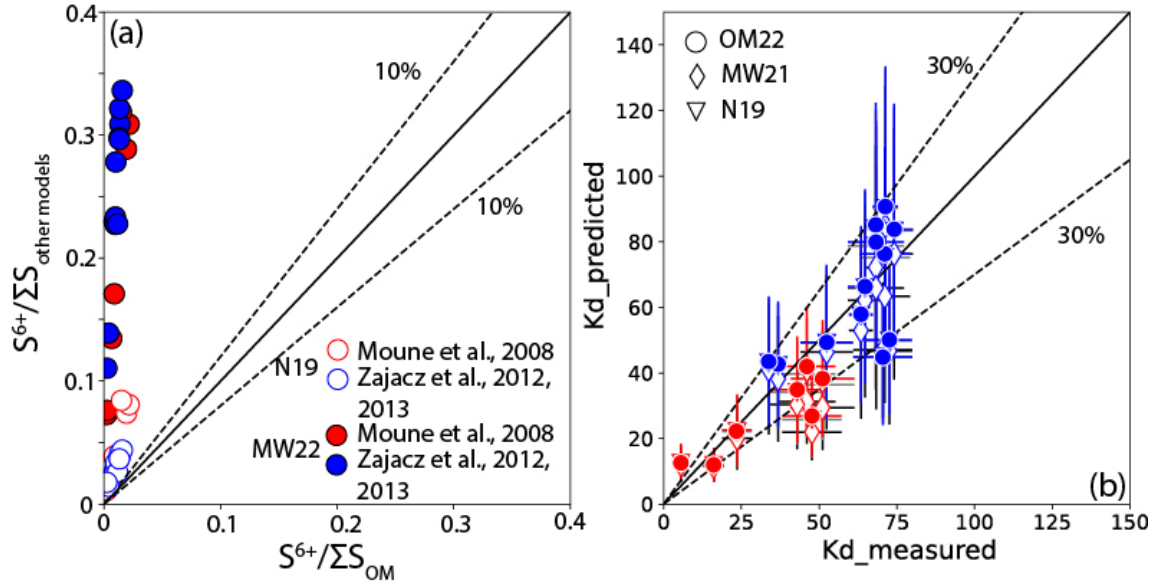

Figure S2. (a) Calculated sulfur speciation using models from O'Neill and Mavrogenes(2022, OM22) compared to results from Muth and Wallace (2020, MW21, filled circles) and Nash et al. (2019, N19, open circles) for experiments without measured $S^{6+}/\Sigma S$. For these experiments, predictions from OM22 and N19 agree well with each other, showing little sulfide in the melt (<10% $S^{6+}/\Sigma S$) while MW21 predicts higher $S^{6+}/\Sigma S$ (up to 35%) in the melt, though still dominated by S^{2-}. (b) Calculated $kd^{v/m}_{combined}$ by OM22, MW21 and N19 compared to the measured $kdS^{v/m}$. Given the differences in predicted $S^{6+}/\Sigma S$, the predicted $kd^{v/m}_{combined}$ from all three models are within 30% of the measured $kdS^{v/m}$. In both panels, red and blue represent experiments from different studies.

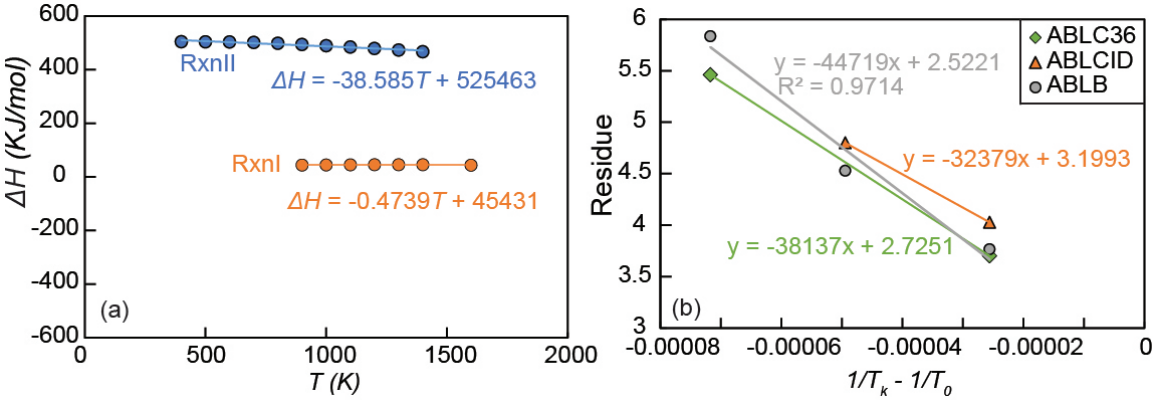

Figure S3. (a) Enthalpy change (ΔH^0_r) at standard state pressure (1atm) of RxnI and RxnII as a function of temperature calculated using the formation enthalpy ($\Delta_f H^0$) from the JANAF table (DOI: 10.18434/T42S31). (b) Empirical fit of the negative enthalpy change ($-\Delta H^0_r$) of RxnII using the temperature series experiments from Fiege et al. (2015).

$\Delta H_r^0 = 32379\ KJ$/mol was employed because it gives the best fit results when regressing for $kdS_{RxnII}^{\frac{vapor}{melt}}$.

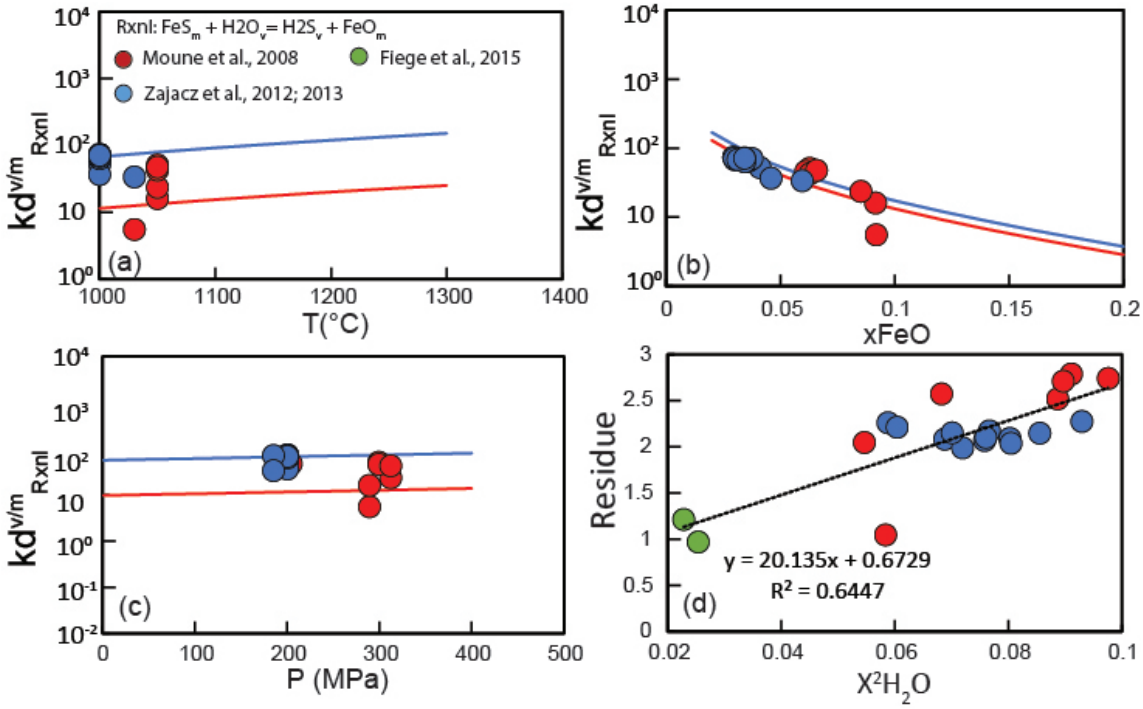

Figure S4. Isolated effect of temperature (a), mole fraction of FeO in the melt (b), and pressure (c) on $kd_{S,RxnI}^{vapor/melt}$. The effect of f_{H_2O} on $kd_{S,RxnI}^{vapor/melt}$ is included in the pressure effect ($f_{H_2O} = P\ \ \varphi_{H_2O,v}\ \ X_{H_2O,v}$). Solid lines are calculated using $kd_{S,RxnI}^{vapor/melt}$ expressions in Table 3 and varying only one variable of interest. Colors of the lines correspond to experimental studies with different melt composition and other conditions. In panel (d), square of the mole fraction of water from RxnI experiments are plotted against the "residue", that is ln ($kd_{S,RxnI}^{\frac{vapor}{melt}}$) corrected for all other factors in Table 3. The dotted line and equation in (d) are calculated from simple linear regression in Excel.

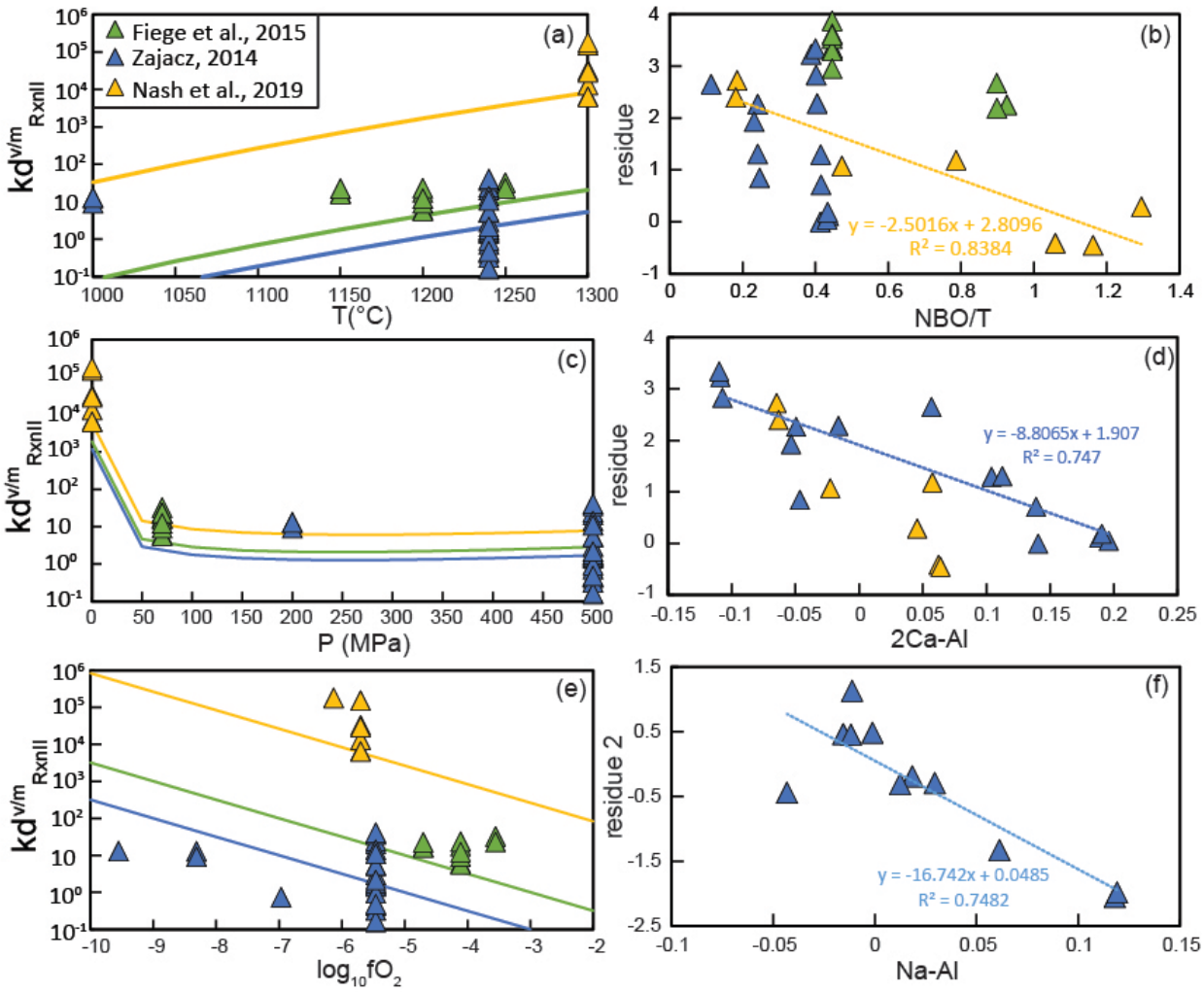

Figure S5. Isolated effects of temperature (a), pressure (c), and fO_2 (e) on $kd_{S,RxnII}^{vapor/melt}$. Solid lines are calculated using the $kd_{S,RxnII}^{vapor/melt}$ expression in Table 3 and varying only one variable of interest. Colors of the lines correspond to the experimental studies with different melt composition and other conditions. In panel (b) and (d), NBO/T and 2Ca-Al of RxnII experiments are plotted against the "residue", that is the ln $(kd_{S,RxnII}^{\frac{vapor}{melt}})$ corrected for *P, T,* and fO_2 effects from the $kd_{S,RxnII}^{vapor/melt}$ expression in Table 3. In panel (f), Na-Al from Na-rich experiments (Zajacz, 2014) that are grouped in RxnII are plotted against the "residue2", that is the ln $(kd_{S,RxnII}^{\frac{vapor}{melt}})$ corrected for P, T, fO_2, NBO/T and 2Ca-Al effects from the expressions in Table 3. The dotted lines and equations in (b), (d), and (f) are calculated from simple linear regression in Excel.

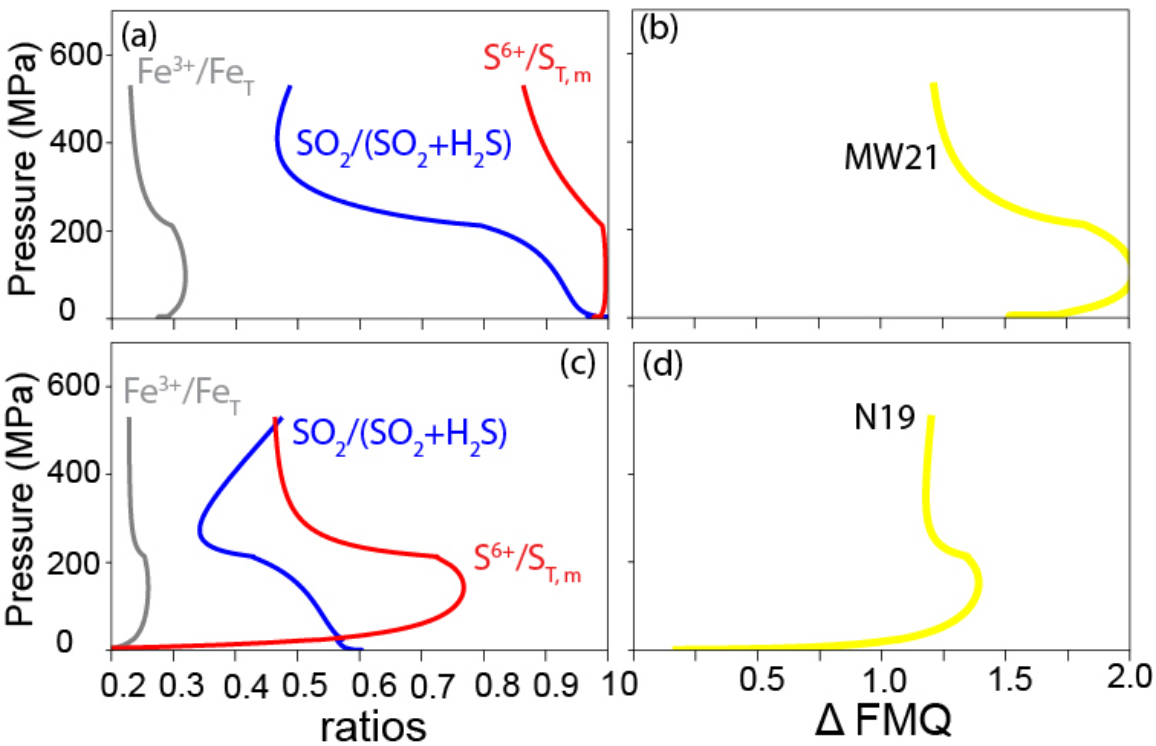

Figure S6. Calculated S speciation in the melt and vapor, Fe speciation in the melt (a, c) and fO_2 relative to FMQ buffer (b, d) using models from Muth and Wallace (2021, MW21) and Nash et al. (2019, N19). Results calculated using OM22 is present in Figure 8 in the maintext.

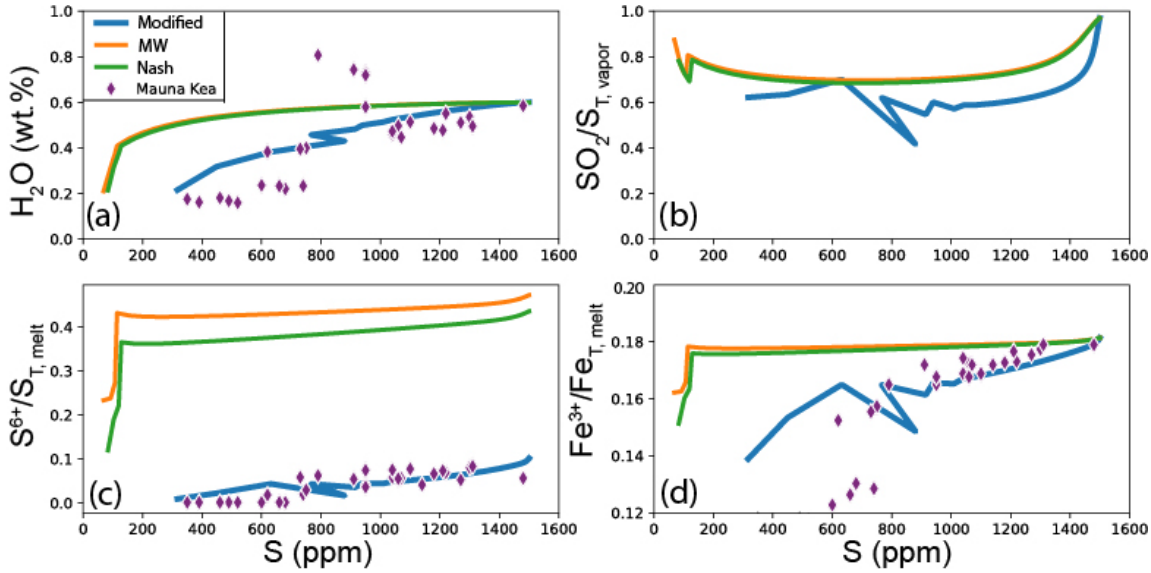

Figure S7. Comparisons of modeled evolution of H_2O concentration in the melt (a), SO_2/S_T in the vapor (b), S^{6+}/S_T in the melt (c) and Fe^{3+}/Fe_T in the melt (d) as S degasses, using the modified model (as described in the main text), Muth and Wallace (2021, MW) and Nash et al. (2019, Nash).

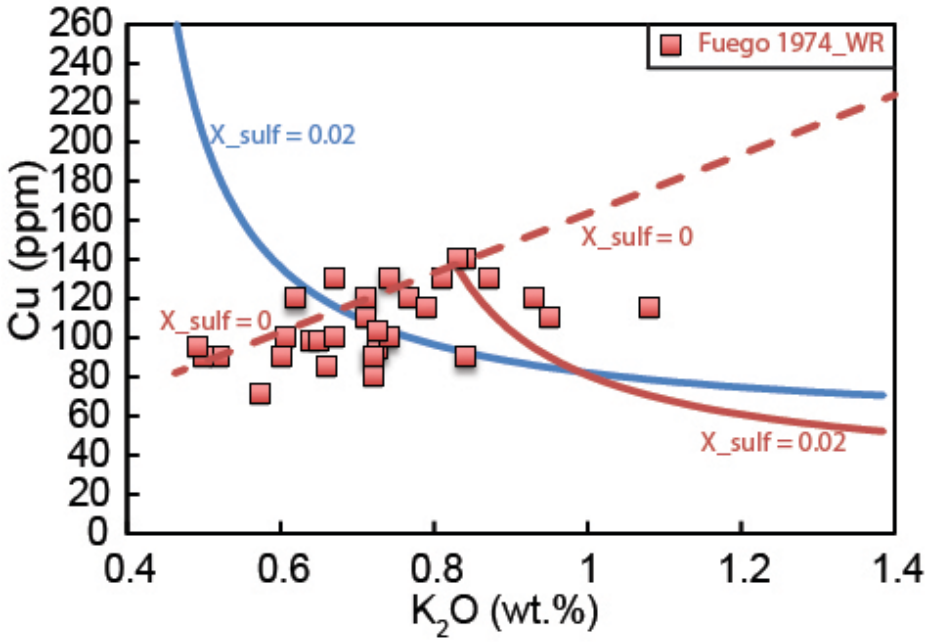

Figure S8. Whole rock Cu vs. K_2O concentration from Fuego 1974 eruption Rose et al. 1978; and Carr and Rose 1987). Red dashed line shows Cu concentration as a function of crystallization (assuming K_2O is perfect incompatible) with no sulfide in the crystal assembly. Blue and red solid curves include 2wt.% sulfide in the crystal assembly from the begining of the crystallization or in the middle of crystallization (K_2O = 0.8wt.%), respectively. Red represents initial Cu of 80 ppm, similar to the whole rock data at low K_2O contents. Blue represents initial Cu of 260 ppm, similar to the highest Cu contents measured from melt inclusions with low K_2O.

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Table S2. Previous experiments on sulfur partition coefficient

Reference	Rxn	Exp.	Exp conditions			Melt															vapor	Partition coefficient			S6+/ST
			P (MPa)	T (°C)	deltaFMQ	SiO2	TiO2	Al2O3	FeO	MnO	MgO	CaO	Na2O	K2O	P2O5	H2O	error	S (ppm)	error	S (wt.%)	kd in wt	error	Kd in mol	method
Moune et	I	B-2	289	1030	-0.1	43.85	4.58	12.69	14.25	0.21	5.42	9.05	2.73	0.55	0.66	4.7	0.47	500		0.711781	14.23561		5.561916	OM22, MW21, N19, Fig. S2a
Moune et	I	B-3	289	1050	-0.1	43.99	4.49	12.6	14.05	0.25	5.38	9.33	2.7	0.53	0.66	4.5	0.45	1360		5.520887	40.59476		16.10344	OM22, MW21, N19, Fig. S2a
Moune et	I	B-13	312	1050	0.1	43.82	4.63	12.71	14.36	0.19	5.43	9.51	2.54	0.59	1.58	6.3	0.63	2370		12.38932	52.27562		23.64766	OM22, MW21, N19, Fig. S2a
Moune et	I	HK-2	299	1050	0.3	51.01	2.01	13.46	10.44	0.32	2.85	6.57	3.43	1.1	1.12	6.3	0.63	300		3.60573	120.191		51.11364	OM22, MW21, N19, Fig. S2a
Moune et	I	HK-3	299	1050	0.3	51.44	1.98	13.66	10.43	0.31	2.89	6.49	3.69	1.15	1.11	6.7	0.67	1250		12.56775	100.542		46.08515	OM22, MW21, N19, Fig. S2a
Moune et	I	HK-6	312	1050	0.1	51.26	2.08	13.68	10.57	0.27	2.9	6.72	3.54	1.12	1.14	6.3	0.63	1350		12.87278	95.3539		42.95249	OM22, MW21, N19, Fig. S2a
Moune et	I	HK-36	204	1050	0.3	52.12	2.12	14.06	10.63	0.24	2.92	6.61	3.71	1.2	1.04	5.3	0.53	400		4.593358	114.834		47.71544	OM22, MW21, N19, Fig. S2a
Zajac et al	I	#145	200	1000	0.2	57	0.52	16.3	6.73	0.16	3.33	6.28	3.24	0.77		5.4	0.4	67	10	0.85	125	20	52.47162	OM22, MW21, N19, Fig. S2a
Zajac et al	I	#149	200	1000	0.2	56.5	0.64	17.4	5.99	0.13	2.97	6.42	3.16	0.78		5.8	0.5	130	11	1.99	153	13	64.78545	OM22, MW21, N19, Fig. S2a
Zajac et al	I	#150	200	1000	0.2	57.5	0.58	17.2	6.05	0.11	3.1	6.37	3.25	0.78		4.9	0.3	228	31	3.9	171	23	70.52393	OM22, MW21, N19, Fig. S2a
Zajac et al	I	#154	200	1000	0.2	57.8	0.55	16.9	5.55	0.06	2.9	6.19	4.04	0.57		5	0.7	227	22	3.97	170	17	72.61471	OM22, MW21, N19, Fig. S2a
Zajac et al	I	#156	200	1000	0.2	57.4	0.57	16.3	5.08	0.12	2.7	5.96	2.37	2.86		6.2	0.4	253	17	3.99	158	8	68.24387	OM22, MW21, N19, Fig. S2a
Zajac et al	I	#159	200	1000	0.2	56.3	0.52	15.3	7.49	0.08	2.82	5.29	3.62	1.69		5.5	0.4	448	33	3.94	88	6	36.81626	OM22, MW21, N19, Fig. S2a
Zajac et al	I	#166	200	1000	0.2	57	0.57	16.7	5.69	0.12	3.36	6.31	3.22	1.48		5.5	0.5	344	20	5.12	149	9	63.4845	OM22, MW21, N19, Fig. S2a
Zajac et al	I	#180	200	1000	0.2	58.3	0.54	17.1	4.95	0.06	3.56	5.38	3.43	0.82		5.8	0.7	239	19	4.12	173	14	74.17146	OM22, MW21, N19, Fig. S2a
Zajac et al	I	#211	200	1000	0.2	56.6	0.64	17.4	5	0.15	2.83	6.57	3.92	0.67		5.8	0.4	247	28	4.09	166	19	71.13102	OM22, MW21, N19, Fig. S2a
Zajac et al	I	#210	200	1000	0.2	57.5	0.56	16.6	5.29	0.09	3.09	6.08	3.79	0.63		6	0.4	259	34	4.09	159	21	68.31129	OM22, MW21, N19, Fig. S2a
Zajac et al	I	#200	185	1000	0	55.86	0.61	17.31	5.92	0.19	2.86	6.48	3.27	0.79		6.6	0.3	354	19	5.664	160	9	71.32591	OM22, MW21, N19, Fig. S2a
Zajac et al	I	#201	185	1030	0	52.1	1.12	14.95	9.87	0.21	3.97	9.24	2.09	0.42		5.9	0.4	670	32	5.293	79	4	33.78899	OM22, MW21, N19, Fig. S2a
Fiege et al	I	ABLCIA-1	70	1050	-0.07	52.46	2.8	16.14		0.15	0.94	10.55	3.31	1.03	0.41	2.73	0.18	231	30	1.1319	49	9	17.82783	XANES
Fiege et al	I	ABLA-1	70	1050	-0.03	52.46	2.8	16.14		0.15	0.94	10.55	3.31	1.03	0.41	2.89	0.29	241	96	1.0363	43	18	15.68864	XANES
Fiege et al	II	ABLB-3	70	1200	4.1	52.46	2.8	16.14	10.98	0.15	0.94	10.55	3.31	1.03	0.41	2.04	0.89	287	14	1.435	50	25	17.93519	XANES
Fiege et al	II	ABLB-2	70	1250	4.1	52.46	2.8	16.14	10.98	0.15	0.94	10.55	3.31	1.03	0.41	2.46	0.34	211	12	1.8568	88	23	32.51766	XANES
Fiege et al	II	ABLB-1	70	1150	4.1	52.46	2.8	16.14	10.98	0.15	0.94	10.55	3.31	1.03	0.41	2.03	0.69	296	36	1.4208	48	20	17.20465	XANES
Fiege et al	II	ABLC36-2	70	1250	4.1	52.46	2.8	16.14	10.98	0.15	0.94	10.55	3.31	1.03	0.41	2.09	0.84	247	31	1.5808	64	29	23.07017	XANES
Fiege et al	II	ABLC36-1	70	1150	4.1	52.46	2.8	16.14	10.98	0.15	0.94	10.55	3.31	1.03	0.41	1.89	0.6	291	28	1.3386	46	17	16.33652	XANES
Fiege et al	II	ABLCID-1	70	1150	4.1	52.46	2.8	16.14	10.98	0.15	0.94	10.55	3.31	1.03	0.41	1.94	0.64	268	45	1.6884	63	26	22.55268	XANES
Fiege et al	II	ABLCID-3	70	1200	4.1	52.46	2.8	16.14	10.98	0.15	0.94	10.55	3.31	1.03	0.41	2.07	0.8	270	15	1.755	65	28	23.46488	XANES
Fiege et al	II	ABWB-3	70	1200	4.1	43.11	2.29	13.96	9.12	0.16	10.6	9.14	2.79	0.84	0.36	2.36	0.09	840	43	1.428	17	1	6.097718	XANES
Fiege et al	II	ABWCIE_3	70	1200	4.1	46.78	2.72	14.87	10.57	0.14	10.52	9.8	2.93	0.92	0.4	2.49	0.09	854	10	1.281	15	1	5.740357	XANES
Fiege et al	II	ABWCI-F3	70	1200	4.1	44.7	2.46	14.69	9.46	0.13	10.34	9.74	2.97	0.92	0.41	2.23	0.09	826	14	2.065	25	3	9.259822	XANES
Fiege et al	II	ABWCID-3	70	1200	4.1	52.46	2.8	16.14	10.98	0.15	0.94	10.55	3.31	1.03	0.41	3.12	0.52	625	33	2	32	11	12.29214	XANES
Zajacz (20	II	#175	200	1000	2.54	56.6	0.52	17.5	5.8	0.14	2.9	6.37	3.2	0.75		6	4	501	29	1.52	30.33932	1.756168	12.91369	Raman
Zajacz (20	II	#165	200	1000	2.54	57.2	0.48	17.1	5.16	0.12	3.33	5.84	3.1	0.73		6.4	4	1901	50	4.01	21.09416	0.554817	9.228318	Raman
Zajacz (20	II	241b	500	1240	2.1	53.76		14.55	0.01		2.36	18.28	0.03	0.01	1.42	9.58	0.3832	5687.769	280.383	3.36	5.907413	0.29121	2.971183	Raman
Zajacz (20	II	244a	500	1240	2.1	53.23		14.38	0.03		0.02	18	3.46	0.01	2.38	8.49	0.3396	9533.022	280.383	2.84	2.979118	0.087621	1.449514	Raman
Zajacz (20	II	245a	500	1240	2.1	51.93		14.55	0.01		0.02	21.76	0.04	0	2.26	9.43	0.3772	9052.365	400.5471	2.98	3.291957	0.145662	1.629812	Raman
Zajacz (20	II	247z	500	1240	2.1	52.34		14.89	0.04		0.01	21.97	0.03		2.29	8.43	0.3372	9172.529	480.6566	2.95	3.216125	0.168531	1.535079	Raman
Zajacz (20	II	248b	500	1240	2.1	52.14		14.69	0.01		0.02	21.85	0.03		2.11	9.15	0.366	8451.545	80.10943	3.02	3.573311	0.03387	1.752342	Raman
Zajacz (20	II	239b	500	1240	2.1	55.16		15.19			4.77	15.77			0.82	8.29	0.3316	3284.487	80.10943	3.6	10.96062	0.267332	5.313388	Raman
Zajacz (20	II	261b	500	1240	2.1	59.01		17.67	0.01		0.02	13.85	0.04	0.01	0.24	9.15	0.366	961.3131	120.1641	3.99	41.50573	5.188216	20.60823	Raman
Zajacz (20	II	262a	500	1240	2.1	56.43		15.88	0.01		0.01	17.05	0.02	0.01	0.87	9.72	0.3888	3484.76	160.2189	3.72	10.67505	0.490807	5.387743	Raman
Zajacz (20	II	245b	500	1240	2.1	56.2		16.08	0.02		16.32	0.19	0.02	0.01	0.22	10.94	0.4376	881.2037	40.05471	3.95	44.82505	2.037502	24.64607	Raman
Zajacz (20	II	238a	500	1240	2.1	56.73		16.2			16.13	0.2			0.072	10.668	0.42672	288.3939	32.04377	2	69.34959	7.705509	36.67773	Raman
Zajacz (20	II	238b	500	1240	2.1	56.55		16.25			16.45	0.19			0.03	10.53	0.4212	120.1641	40.05471	0.95	79.05853	26.35284	41.02421	Raman
Zajacz (20	II	239a	500	1240	2.1	56.17		15.64			10.99	7.37			0.35	9.48	0.3792	1401.915	40.05471	3.87	27.6051	0.788717	14.24325	Raman
Zajacz (20	II	263a	500	1240	2.1	52.61		14.42	0.02		0.03	16.82	5.45	0.01	3.29	7.35		13178	480.6566	2.13	1.61633	0.058954	0.73946	Raman
Zajacz (20	II	258a	500	1240	2.1	50.67		13.6	0.01		0.02	13.95	6.94	0.02	4.49	10.3		17984.57	841.149	1.6	0.889652	0.04161	0.454366	Raman
Zajacz (20	II	263b	500	1240	2.1	53.68		14.59	0		0.02	14.44	7.93	0.01	1.55	7.78		6208.481	240.3283	0.59	0.950313	0.036786	0.44417	Raman
Zajacz (20	II	242a	500	1240	2.1	54.8		14.68	0.05		9.15	0.19	8.82	0.07	2.09	10.15		8371.435	360.4924	2.91	3.476106	0.149689	1.82996	Raman
Zajacz (20	II	244b	500	1240	2.1	51.07		13.76	0.02		0.02	9.86	10.86	0.02	4.89	9.5		19586.76	1241.696	1.27	0.648397	0.041105	0.328614	Raman
Zajacz (20	II	258b	500	1240	2.1	53.18		14.02	0		6.36	0.35	10.26	0.4	3.18	12.25		12737.4	1281.751	2.14	1.680092	0.169066	0.942843	Raman
Zajacz (20	II	250a	500	1240	2.1	51.2		13.73	0.03		0.02	0.04	18.92	0.14	5.87	10.05		23512.12	1602.189	0.7	0.297719	0.020287	0.159625	Raman
Zajacz (20	II	255b	500	1240	2.1	52.24		13.85	0.01		0.02	0.02	19.59	0.09	3.11	11.07		12457.02	881.2037	0.38	0.305049	0.021579	0.168769	Raman
Zajacz (20	II	251b	500	1240	2.1	58.72		16.9	0.02		0.02	0.01	11.39	0.03	1.98	10.93		7930.833	921.2584	3.21	4.047494	0.470163	2.166986	Raman
Zajacz (20	II	250b	500	1240	2.1	53.38		14.93	0.02		0.03	0.04	14.84	0.09	4.57	12.1		18305	1441.97	1.58	0.863152	0.067994	0.484846	Raman
Zajacz (20	II	279b	500	1240	2.1	61		17.79	0.01		0.02	0.02	9.81	0.01	0.46	10.88	0.4352	1842.517	160.2189	3.69	20.02695	1.741474	11.04321	Raman
Nash et al.	II	JFR+1.60	0.1	1300	1.519418	45.84	1.46	12.46	17.07	0.39	8.31	9.88	2.44	0.04		0		84	17	50.04995	5958.327		6607.709	XANES
Nash et al.	II	DWF+1.60	0.1	1300	1.519418	49.89	0.41	7.73	20.86	0.58	12.06	6.43	0.18		0.04	0		41	19	50.04995	12207.3		13340.83	XANES
Nash et al.	II	PRI+1.60	0.1	1300	1.519418	48.12	1.23	11.71	12.3	0.17	15.2	9.61	0.05			0		89.3	83	50.04995	5604.697		6355.785	XANES
Nash et al.	II	EVO+1.60	0.1	1300	1.519418	51.92	3.71	10.02	19.39	0.39	4.48	8.16	0.11			0		16.1	53	50.04995	31086.92		32719.34	XANES
Nash et al.	II	AND+1.60	0.1	1300	1.519418	53.97	1.32	16.55	10.39	0.15	4.31	7.99	3.24	0.68		0		19.3	3	50.04995	25932.62		29152.2	XANES
Nash et al.	II	DAC+1.60	0.1	1300	1.519418	65.6	0.67	16.3	5.36	0.11	2.15	5.75	3.03	1		0		3.73	20	50.04995	134182.2		152249.3	XANES
Nash et al.	II	DAC+1.21	0.1	1300	1.089418	64.95	0.68	16.46	4.4	0.12	2.22	5.92	3.37	1.12		0		1.49	9	21.30405	142980.2		179779.8	XANES